DHANASHREE SOMANI

PhD Candidate in Statistics

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PROFESSIONAL SUMMARY

PhD candidate in Statistics with expertise in time series econometrics, stochastic processes, and quantitative modeling. Skilled in Python, R, and C++ for forecasting, risk modeling, and financial data analysis. Published in finance research letters, with experience applying machine learning and statistical inference to market volatility and macroeconomic indicators.

TECHNICAL SKILLS

- **Quantitative Research & Finance**:
  - Derivatives Pricing**:** Black Scholes, Binomial and Trinomial Trees, Heston model, SABR Model.
  - Volatility modelling: GARCH and variants, Stochastic vs Historical Volatility comparison.
  - Other Models: ARIMA, SARIMA, VAR, Machine-Learning (XGBoost, LSTM, BNN, etc), Time Series regime detection.
- **Programming**: Python, R, MATLAB, C++, Java, LaTeX.
- **Statistics & Mathematics**: Time series modelling and analysis, Hypothesis testing, Bayesian Analysis, GLMs, Optimization methods, Probability Theory, Numerical Analysis, Linear Algebra.

EDUCATION

UNIVERSITY OF FLORIDA — GAINESVILLE, FLORIDA

PhD in Statistics — 2022-Present

- **PhD Thesis:** Exploring comparative forecasting performance of different sequences of nested models, to understand whether a parsimonious model can yield forecasts of statistically equal accuracy, via different econometric metrics; Transfer Learning for Forecasting the Current Regime in Piecewise-Constant High-Dimensional VAR Models.

UNIVERSITY OF FLORIDA — GAINESVILLE, FLORIDA

MS in Statistics — 2022-2024

- **MS projects**: Implementation and extension of the well-known Clark-West Test for Predictive Accuracy in nested models, specifically applied to INGARCH models
- **Relevant Courses:** Measure and Probability Theory, Regression Analysis, Inference, Optimization, Markov chain Monte Carlo Methods, Survival Analysis.

INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH — KOLKATA, INDIA

MS in Mathematics and Statistics (BS-MS Dual Degree Course; INSPIRE Fellow) — 2017-2022

- **MS Thesis:** Drift Diffusion Model in the TAFC (Two-alternative forced choice) Paradigm and Bayesian Inference
- **Relevant Courses:** Numerical Analysis, Econometrics, Machine Learning and Network analysis, Linear Algebra, Graph theory, Number theory.
- **Summer Projects**
  - Jump discontinuities in financial time series data and stochastic calculus
  - Random Graph theory and network analysis

PROJECTS AND PUBLICATIONS

PUBLICATIONS:

- Shortages and machine-learning forecasting of oil returns volatility: 1900–2024, *Finance Research Letters, Volume 79, 2025.*
  The objective of this paper is to forecast the volatility of the West Texas Intermediate oil returns over the monthly period of January 1900 to June 2024 by utilizing the information content of newspapers articles-based indexes shortages for the United States.
- Supply bottlenecks and machine-learning forecasting of international stock market volatility, *Finance Research Letters, Volume 86, Part G, 2025.*
  This study explores the information value of the daily Supply Bottlenecks Index – derived from newspaper articles – to forecast daily return volatilities of stock markets of seven major economies – China, France, Germany, Italy, Spain, the UK, and the US.

PROJECTS IN QUANT RESEARCH:

- Dynamic Volatility Modelling for Option Hedging: Designed rolling-window backtests across five major equities to evaluate the delta-hedging performance of dynamic volatility models (GARCH, GJR-GARCH) against Black-Scholes baselines.

CERTIFICATIONS AND OTHER COURSES

- Quant Finance Boot Camp: Certificate Program by the Erdős Institute.

PROFESSIONAL EXPERIENCE

UNIVERSITY OF FLORIDA — GAINESVILLE, FLORIDA

Instructor of Record — 2023-Present

Designed and taught Engineering Statistics Course for classes of 70+ undergraduate students over 2 semesters. It is a foundational statistics course covering topics like probability, discrete and continuous random variables, confidence interval estimation, hypothesis testing, correlation, regression, and analysis of variance.

Graduate Teaching Assistant — 2022-Present

Teaching Assistant for various undergraduate courses like Statistical methods in research, Multivariate statistical analysis, Design of experiments, Statistical modelling, etc.

EXTRA-CURRICULAR

- Reviewer for journals like Sankhya A, Finance Research Open.
- Received DST-SHE INSPIRE scholarship (2017) - offered to top 1 percentile students in the boards.